

Пример 11.

Коды Руга-Саммонса

БЧХ-коды поля.

α
↑

$$g(x) = \text{НОК} \{ M_i(x), i = 0, \dots, b+d-2 \}$$

$$g_0 + g_1 x + g_2 x^2 + \dots + g_m x^m$$

$$G = \begin{pmatrix} g_0 & g_1 & g_2 & \dots & g_m \\ & g_0 & g_1 & \dots & g_m \\ & & g_0 & g_1 & \dots & g_m \\ & & & g_0 & g_1 & \dots & g_m \\ & & & & g_0 & g_1 & \dots & g_m \end{pmatrix}$$

$k < n$
 $n = n - k$

$$GF(2) = \{1, 1\}$$

$$GF(p) = a_0 + a_1 x + a_2 x^2 + \dots \quad a_i \in GF(p)$$

$$p(x) = \deg p(x) = m$$

$$1, \alpha, \alpha^2, \dots, \alpha \rightarrow M_2(x)$$

$$x - \alpha = \emptyset$$

$$x - 1 = \emptyset$$

Коды PC над полем $GF(p)$ код БЧХ
 $n = p - 1$

$$g(x) = \prod_{b=1}^{b+d-2} (x - \alpha^b) \dots (x - \alpha^{b+d-2})$$

$$n = 15 \quad GF(2^n) \quad d = 5$$

$$g(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^3)(x - \alpha^4) =$$

$$x^4 + \alpha_3 x^3 + \alpha_2 x^2 + \alpha_1 x + \alpha_0$$

$$g(x) \quad k = 11 \quad n = 15 \quad d = 5 \quad \begin{pmatrix} 15, 2 \\ 15, 11 \end{pmatrix}$$

Алгоритм Рунге-Кутты - Горюхиной-Суряева

$GF(q)$ д $\alpha, \alpha^2, \dots, \alpha^{d-1}$ - корни $f(x)$

$$c(x) = c_0 + c_1 x + \dots + c_{n-1} x^{n-1} \quad t = \left\lfloor \frac{d-1}{2} \right\rfloor$$

$$e(x) = e_0 + e_1 x + \dots + e_{n-1} x^{n-1}$$

$$r(x) = c(x) + e(x) = v_0 + v_1 x + \dots + v_{n-1} x^{n-1}$$

$$c(\alpha^j) = \neq, \quad j = \overline{1, d-1}$$

$$s_j = \sum_{v < t} v(\alpha^j) = c(\alpha^j) + e(\alpha^j) = e(\alpha^j)$$

$i_1, i_2, \dots, i_r$ - ошибки

$$s_j = e_{i_1} \cdot \alpha^{i_1 j} + e_{i_2} \cdot \alpha^{i_2 j} + \dots + e_{i_r} \cdot \alpha^{i_r j}$$

здесь ошибки i_2

$$y_h = e_{i_h} \quad X_h = \alpha^{i_h} \quad \text{номер ошибки}$$

$$s_j = y_1 \cdot X_1^j + \dots + y_r \cdot X_r^j \quad j = \overline{1, d-1}$$

гроз
ошибки

Если $d < t$

Вместо X_h будем искать корни многочлена, корни ког. двл X_h

$$\Delta(x) = \prod_{j=1}^r (1 - x \cdot X_j) = \text{полный многочл ошибок}$$

$$= 1 + \Delta_1 x + \dots + \Delta_r x^r$$

$$d=3 \quad y_1 \cdot X_1 + y_2 \cdot X_2 + y_3 \cdot X_3 = S_1$$

$$y_1 \cdot X_1^2 + y_2 \cdot X_2^2 + y_3 \cdot X_3^2 = S_2$$

$$y_1 \cdot X_1^3 + y_2 \cdot X_2^3 + y_3 \cdot X_3^3 = S_3$$

$$\Delta(X_h^{-1}) = 1 + \Delta_1 X_h^{-1} + \Delta_2 X_h^{-2} + \Delta_3 X_h^{-3} = \neq$$

$$\begin{cases} \Delta(x_2^{-1}) = 1 + \Delta_1 x_2^{-1} + \Delta_2 x_2^{-2} + \Delta_3 x_2^{-3} = \phi \\ \Delta(x_3^{-1}) = 1 + \Delta_1 x_3^{-1} + \Delta_2 x_3^{-2} + \Delta_3 x_3^{-3} = \phi \end{cases}$$

$$\begin{cases} x_1^3 + \Delta_1 x_1^2 + \Delta_2 x_1 + \Delta_3 = \phi \\ x_2^3 + \Delta_1 x_2^2 + \Delta_2 x_2 + \Delta_3 = \phi \\ x_3^3 + \Delta_1 x_3^2 + \Delta_2 x_3 + \Delta_3 = \phi \end{cases}$$

$$\begin{aligned} x_1^4 y_1 + \Delta_1 x_1^3 y_1 + \Delta_2 x_1^2 y_1 + \Delta_3 x_1 y_1 &= 0 \\ x_2^4 y_2 + \Delta_1 x_2^3 y_2 + \Delta_2 x_2^2 y_2 + \Delta_3 x_2 y_2 &= 0 \\ x_3^4 y_3 + \Delta_1 x_3^3 y_3 + \Delta_2 x_3^2 y_3 + \Delta_3 x_3 y_3 &= 0 \end{aligned}$$

$S_4 \qquad S_3 \qquad S_2 \qquad S_1$

$$S_4 + \Delta_1 S_3 + \Delta_2 S_2 + \Delta_3 S_1 = \phi$$

$$\rightarrow S_5 + \Delta_1 S_4 + \Delta_2 S_3 + \Delta_3 S_2 = \phi$$

$$\rightarrow S_6 + \Delta_1 S_5 + \Delta_2 S_4 + \Delta_3 S_3 = \phi$$

$$\begin{pmatrix} S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \\ S_3 & S_4 & S_5 \end{pmatrix} \begin{pmatrix} \Delta_3 \\ \Delta_2 \\ \Delta_1 \end{pmatrix} = \begin{pmatrix} -S_4 \\ -S_5 \\ -S_6 \end{pmatrix}$$

$$GF(2^7) \quad PC(15, 9) \quad d=3$$

$$p(x) = 1 + x + x^7$$

$$\begin{aligned} g(x) &= (x + \alpha)(x + \alpha^2)(x + \alpha^3)(x + \alpha^4)(x + \alpha^5)(x + \alpha^6) \\ &= x^6 + x^5(\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) + \dots \end{aligned}$$

$$\alpha + \alpha^2 + \cancel{\alpha^3} + (1 + \cancel{\alpha}) + (\cancel{\alpha} + \alpha^2) + (\cancel{\alpha^3} + \cancel{\alpha^4}) =$$

$$= 1 + \alpha + \alpha^2 = \alpha^{10}$$

$$g(x) = x^6 + \alpha^{10} + \alpha^5 + \dots$$

$$c(x) = \alpha + \alpha^3 x + \dots$$

$$x^{14}$$

$$s(x) =$$

$$e(x) = ?$$